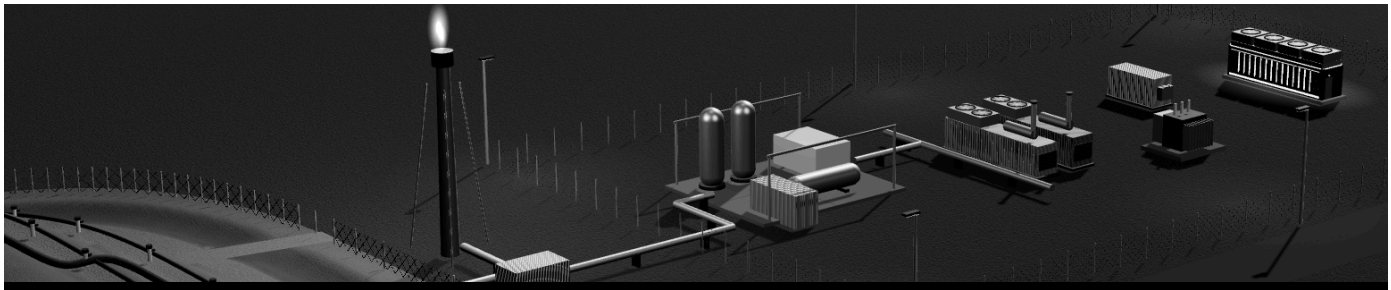
# City of Richmond Wastewater Treatment Plant, Richmond, Virginia: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

| 1.0 MW | 1 | 6 ac | 91k |
|---|---|---|---|
| POWER FROM THE GAS | COMPUTE MODULES | LAND NEEDED | MMBTU BOUGHT / YR |

| GAS SALE, PER YEAR | LAND LEASE, PER YEAR | TOTAL TO THE OWNER, PER YEAR |
|---|---|---|
| up to $68k | $6k to $12k | $6k to $81k |



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=300&CH4=60

## THE MATH

| | |
|---|---|
| Digester gas, low estimate | 300 scfm |
| Methane (assumed) | 60% |
| Energy in the gas | 10.9 MMBtu per hour |
| Power at a heat rate of 11,000 Btu/kWh | about 1.0 MW |
| Compute (power less cooling and losses, /1.2) | about 0.8 MW |
| Compute modules (about 1 MW each) | 1 |
| Gas engines (running plus one spare) | 2 at 1.6 MW |
| Land: 5 acres plus 1.2 acres per MW | about 6 acres |
| Gas bought per year (95% run time) | 91k MMBtu |
| Gas payment, up to $0.75/MMBtu | up to $68k |
| Land lease at $1,000 to $2,000 per acre per year | $6k to $12k |

Biogas: methane at about 1,012 Btu per cubic foot. Prices in US dollars; the price band is the one shown on our Site model page.

## WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

### WORTH CHECKING

- Flow is an estimate, not a measurement
- How much gas is flared or used for heat today
- Space for a parcel of one to two acres

**Source:** No gas figure published. Estimate: about 6.9 scfm of digester gas per MGD (EPA CHP Partnership rule of thumb) at about 61.6 MGD gives about 425 scfm; this first read takes about 300 scfm, leaving the rest for digester heat. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.